

## Lecture 2

### Span, Linear Independence, Basis, Dimension

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Throughout,  $\mathbb{F}$  is  $\mathbb{R}$  or  $\mathbb{C}$ . Tags indicate the flavour of each problem; a ★ marks a harder one. Problems that ask whether a vector lies in a span or whether vectors are linearly independent, and problems that ask for the coordinates of a vector, all reduce to solving linear systems.

#### Core tutorial route

Work **2.1, 2.12, 2.13, 2.15, 2.17, 2.22, 2.26, and 2.27** in that order. This route covers span, field-sensitive independence, basis construction and extension, the linear-dependence lemma, coordinate calculation and proof, and a dimension-formula transfer.

**Additional practice:** 2.2–2.11, 2.14, 2.16, 2.18–2.21, 2.23–2.25, 2.28–2.30. **Bridge:** none required. **Extension:** 2.31.

#### Span

**Exercise 2.1** (computational) Is  $(1, 2, 3)$  in  $\text{span}((1, 1, 0), (0, 1, 1))$  in  $\mathbb{R}^3$ ?

Answer. No: the system is inconsistent, so  $(1, 2, 3) \notin \text{span}((1, 1, 0), (0, 1, 1))$ .

**Exercise 2.2** (computational) Is  $(2, 3, 1)$  in  $\text{span}((1, 1, 0), (0, 1, 1))$  in  $\mathbb{R}^3$ ? If so, write it as a linear combination.

Answer. Yes:  $(2, 3, 1) = 2(1, 1, 0) + (0, 1, 1)$ .

**Exercise 2.3** (computational) For which value of  $t$  is  $(t, 1, 2)$  in  $\text{span}((1, 0, 1), (1, 1, 1))$  in  $\mathbb{R}^3$ ?

Answer.  $t = 2$ .

**Exercise 2.4** (computational) Does the polynomial  $p(t) = 2 + 5t + 3t^2 \in \mathcal{P}_2(\mathbb{R})$  lie in the subspace  $\text{span}(1 + t, t + t^2)$ ?

Answer. Yes:  $2 + 5t + 3t^2 = 2(1 + t) + 3(t + t^2)$ .

**Exercise 2.5** (proof) Show that  $\text{span}(v_1, \dots, v_m) = \text{span}(v_1, \dots, v_m, w)$  if and only if  $w \in \text{span}(v_1, \dots, v_m)$ .

Hint. One direction is immediate. For the other, substitute  $w = \sum a_i v_i$  into a combination that uses  $w$  and collect the  $v_i$ .

#### Linear independence

**Exercise 2.6** (computational) Find the value of  $t$  for which  $(3, 1, 4), (2, -3, 5), (5, 9, t)$  are linearly dependent in  $\mathbb{R}^3$ .

Answer.  $t = 2$ .

**Exercise 2.7** (computational) Are  $(1, 3), (2, 5)$  linearly independent in  $\mathbb{R}^2$ ?

Answer. Yes.

**Exercise 2.8** (computational) Are  $(1, 0, 1)$ ,  $(2, 1, 0)$ ,  $(3, 1, 2)$  linearly independent in  $\mathbb{R}^3$ ?

Answer. Yes (the determinant is 1).

**Exercise 2.9** (computational) Are  $1 + t$ ,  $1 - t$ ,  $t^2$  linearly independent in  $\mathcal{P}_2(\mathbb{R})$ ?

Answer. Yes.

**Exercise 2.10** (computational) Are  $(1, i)$ ,  $(i, -1)$  linearly independent in  $\mathbb{C}^2$ ?

Answer. No: they are linearly dependent.

**Exercise 2.11** (proof) Suppose  $v_1, v_2, v_3, v_4$  are linearly independent in  $V$ . Prove that

$$v_1 - v_2, \quad v_2 - v_3, \quad v_3 - v_4, \quad v_4$$

are also linearly independent.

**Hint.** Set a general combination equal to 0 and gather the coefficient of each  $v_i$ ; independence of the original list gives a system you can solve one variable at a time.

**Exercise 2.12** (★) View  $\mathbb{C}$  as a vector space. Show that  $1 + i$ ,  $1 - i$  are linearly independent over  $\mathbb{R}$  but linearly dependent over  $\mathbb{C}$ .

**Hint.** The scalar field is doing the work. Over  $\mathbb{R}$ , separate real and imaginary parts; over  $\mathbb{C}$ , ask what multiplying  $1 + i$  by  $i$  gives.

## Bases and dimension

**Exercise 2.13** (computational) Find a basis and the dimension of  $U = \{x \in \mathbb{R}^4 : x_1 - x_2 + 2x_3 = 0, x_2 - x_3 + x_4 = 0\}$ .

Answer. Basis  $(-1, 1, 1, 0)$ ,  $(-1, -1, 0, 1)$ ;  $\dim U = 2$  (any basis of  $U$  is acceptable).

**Exercise 2.14** (computational) Give a basis and the dimension of (a) the symmetric  $2 \times 2$  real matrices, and (b) the trace-zero  $2 \times 2$  real matrices.

Answer. (a) dimension 3; (b) dimension 3.

**Exercise 2.15** (computational) Extend the independent list  $(1, 2, 1, 0)$ ,  $(0, 1, 1, 1)$  to a basis of  $\mathbb{R}^4$ .

Answer. E.g.  $(1, 2, 1, 0)$ ,  $(0, 1, 1, 1)$ ,  $(1, 0, 0, 0)$ ,  $(0, 0, 1, 0)$  (other choices work).

**Exercise 2.16** (computational) Find a basis and the dimension of  $U = \{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$ .

Answer. Basis  $(-1, 1, 0)$ ,  $(-1, 0, 1)$ ;  $\dim U = 2$ .

**Exercise 2.17** (computational) Find a basis and the dimension of  $\{p \in \mathcal{P}_3(\mathbb{R}) : p(0) = 0\}$ .

Answer. Basis  $t$ ,  $t^2$ ,  $t^3$ ; dimension 3.

**Exercise 2.18** (computational) Find a basis and the dimension of the space  $U$  of diagonal  $3 \times 3$  real matrices.

Answer. Basis  $\text{diag}(1, 0, 0)$ ,  $\text{diag}(0, 1, 0)$ ,  $\text{diag}(0, 0, 1)$ ;  $\dim U = 3$ .

**Exercise 2.19** (computational) Use the linear-dependence lemma to remove a redundant vector, then find the dimension of the subspace

$$U = \text{span}((1, 2, 3, 4), (2, 4, 6, 8), (1, 0, 1, 0), (0, 1, 0, 1)) \subseteq \mathbb{R}^4.$$

Answer. 3.

**Exercise 2.20** (proof) Let  $V$  be finite-dimensional and let  $U \subseteq V$  be a subspace with  $\dim U = \dim V$ . Prove that  $U = V$ .

**Hint.** Take a basis of  $U$  and count: it is an independent list of length  $\dim V$  sitting inside  $V$ .

**Exercise 2.21** (proof) Let  $u_1, u_2, u_3$  be a basis of a vector space  $V$ . Prove that the vectors  $u_1, u_1 + u_2, u_1 + u_2 + u_3$  also form a basis of  $V$ .

**Hint.** Set a combination of the three new vectors equal to 0 and collect the coefficients of  $u_1, u_2, u_3$ ; the resulting system is triangular, so solve it from the bottom up. Then count the list.

## Coordinates

**Exercise 2.22** (computational) Find the coordinate vector  $[v]_{\mathcal{B}}$  of  $v = (1, 2, 3)$  in the basis  $\mathcal{B} = ((1, 1, 0), (0, 1, 1), (0, 0, 1))$ .

Answer.  $[v]_{\mathcal{B}} = (1, 1, 2)$ .

**Exercise 2.23** (computational) Find the coordinate vector  $[v]_{\mathcal{B}}$  of  $v = (5, 1)$  in the basis  $\mathcal{B} = ((1, 1), (1, -1))$ .

Answer.  $[v]_{\mathcal{B}} = (3, 2)$ .

**Exercise 2.24** (computational) Find the coordinate vector  $[p]_{\mathcal{C}}$  of  $p(t) = t^2 \in \mathcal{P}_2(\mathbb{R})$  in the basis  $\mathcal{C} = (1, t - 1, (t - 1)^2)$ .

Answer.  $[p]_{\mathcal{C}} = (1, 2, 1)$ .

**Exercise 2.25** (computational) Find the coordinate vector  $[v]_{\mathcal{B}}$  of  $v = (2, 3, 4)$  in the basis  $\mathcal{B} = ((1, 0, 0), (1, 1, 0), (1, 1, 1))$ .

Answer.  $[v]_{\mathcal{B}} = (-1, -1, 4)$ .

**Exercise 2.26** (proof) Let  $\mathcal{B}$  be a basis of an  $n$ -dimensional space  $V$ . Prove that vectors  $v_1, \dots, v_k \in V$  are linearly independent if and only if their coordinate vectors  $[v_1]_{\mathcal{B}}, \dots, [v_k]_{\mathcal{B}}$  are linearly independent in  $\mathbb{F}^n$ .

**Hint.** The coordinate map is a bijection that respects sums and scalar multiples, so it carries a relation among the  $v_i$  to a relation among their coordinate vectors, and back.

## Sums and dimension formula

**Exercise 2.27** (computational) In  $\mathbb{R}^4$  let

$$U = \text{span}((1, 1, 1, 1), (1, 0, 1, 0)) \quad \text{and} \quad W = \text{span}((0, 1, 0, 1), (1, 1, 0, 0)).$$

Find  $\dim(U + W)$  and  $\dim(U \cap W)$ , and exhibit a basis of  $U \cap W$ .

Answer.  $\dim(U + W) = 3$ ,  $\dim(U \cap W) = 1$ ,  $U \cap W = \text{span}((0, 1, 0, 1))$ .

**Exercise 2.28** (computational) In  $\mathbb{R}^3$  let

$$U = \text{span}((1, 0, 0), (0, 1, 0)) \quad \text{and} \quad W = \text{span}((0, 1, 0), (0, 0, 1)).$$

Find  $\dim(U + W)$  and  $\dim(U \cap W)$ , and give a basis of  $U \cap W$ .

**Answer.**  $\dim(U + W) = 3$ ,  $\dim(U \cap W) = 1$ ,  $U \cap W = \text{span}((0, 1, 0))$ .

**Exercise 2.29** (computational) In  $\mathbb{R}^4$  let

$$U = \text{span}((1, 0, 0, 0), (0, 1, 0, 0)) \quad \text{and} \quad W = \text{span}((0, 0, 1, 0), (0, 0, 0, 1)).$$

Find  $\dim(U + W)$  and  $\dim(U \cap W)$ . Is the sum direct?

**Answer.**  $\dim(U + W) = 4$ ,  $\dim(U \cap W) = 0$ ; the sum is direct.

**Exercise 2.30** (computational) In  $\mathbb{R}^4$  let

$$U = \text{span}((1, 1, 0, 0), (0, 0, 1, 1)) \quad \text{and} \quad W = \text{span}((1, 0, 0, 1), (0, 1, 1, 0)).$$

Find  $\dim(U + W)$  and  $\dim(U \cap W)$ , and exhibit a basis of  $U \cap W$ .

**Answer.**  $\dim(U + W) = 3$ ,  $\dim(U \cap W) = 1$ ,  $U \cap W = \text{span}((1, 1, 1, 1))$ .

**Exercise 2.31** (★) Let  $V$  be finite-dimensional with  $\dim V = n$ , and let  $U, W$  be subspaces with  $\dim U + \dim W > n$ . Prove that  $U \cap W \neq \{0\}$ .

**Hint.** Apply the dimension formula to  $U + W$ , and use that  $U + W$  is still a subspace of the  $n$ -dimensional  $V$ .